

Candidate Name

Centre Number

Candidate Number



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

CHEMISTRY**9189/2**

PAPER 2 Theory

JUNE 2016 SESSION**1 hour 15 minutes**

Candidates answer on the question paper.

Additional materials:

Data Booklet

Mathematical Tables and/or Calculator

TIME 1 hour 15 minutes**INSTRUCTIONS TO CANDIDATES**

Write your name, Centre number and candidate number in the spaces at the top of this page.

Answer **all** questions.

Write your answers in the spaces provided on the question paper.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

FOR EXAMINER'S USE

1	
2	
3	
4	
5	
TOTAL	

This question paper consists of 8 printed pages.

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- 1 The element, R, has components 31x and 31y proportions x% and y % respectively.

(i) Write the electronic configuration of R.

(ii) State the name given to the two components of R.

(iii) Calculate the % proportions x and y if the A_r of R is 69.8.

[Total: 6]

- 2 (a) The reaction between trimethylamine, $N(CH_3)_3$, and boron trifluoride, BF_3 , forms a solid salt.

(i) Draw the structural formula to show bonding in the salt.

- (ii) Name the bond formed between the two compounds.

_____ [3]

- (b) The pH of 0.1 M hydrochloric acid is 1 and that of 0.1 M ethanoic acid is approximately 3.

- (i) Write the equation for the dissociation of ethanoic acid.

- (ii) Explain the difference in pH of the two acids.

_____ [3]

- (c) 0.1 M ammonia and 0.1 M ammonium chloride are mixed to give a buffer solution of pH 10.0

[K_a for ammonium chloride is $6 \times 10^{-10} \text{ mol dm}^{-3}$]

- (i) Define the term *buffer solution*.

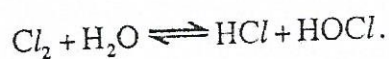
- (ii) Calculate the ratio $\frac{[\text{NH}_4^+]}{[\text{NH}_3]}$ in the buffer solution.

[4]

[Total: 10]

3

- (a) Chlorine reacts with water to form a green solution that smells chlorine. The smell of chlorine disappears when an alkali is added to the solution. The chemical equation for the reaction of chlorine with water is



Suggest why

- (i) the solution smells chlorine,

- (ii) the addition of aqueous alkali removes the smell of chlorine.

[4]

- (b) Table 2 shows melting points of two Group (IV) elements.

Table 2

element	melting points/ $^{\circ}\text{C}$
C(diamond)	3 730
Pb	327

- (i) Explain the difference in the melting points of the elements in terms of structure and bonding.

- (ii) Write the formula of the stable oxides of

1. lead, _____
2. diamond. _____

- (iii) State the acid-base nature of the solutions of the two oxides when reacted with water.

- (iv) Describe and explain the trend in the +4 oxidation state down the Group (IV) oxides.

[8]

[Total: 12]

- 4 (a) Compounds Q and R are isomers of M_r 72 and each contains 66.6% carbon, 11.2% hydrogen and 22.2% oxygen.

Calculate the

- (i) empirical formula of Q,

- (ii) molecular formula of Q.

[3]

- (b) Q and R give a yellow precipitate with 2,4-dinitrophenylhydrazine. Q gives a red brown precipitate with Fehling's solution and R gives a yellow precipitate with triiodomethane.

Deduce, with reasons, the structural formula of

- (i) Q,

reason _____

- (ii) R.

reason _____

[3]

- (c) (i) Name and draw the structure of the organic product formed when R reacts with sodium tetrahydroborate (III)

name _____

structure _____

- (ii) Name the type of reaction in (i).

[6]

[Total: 12]

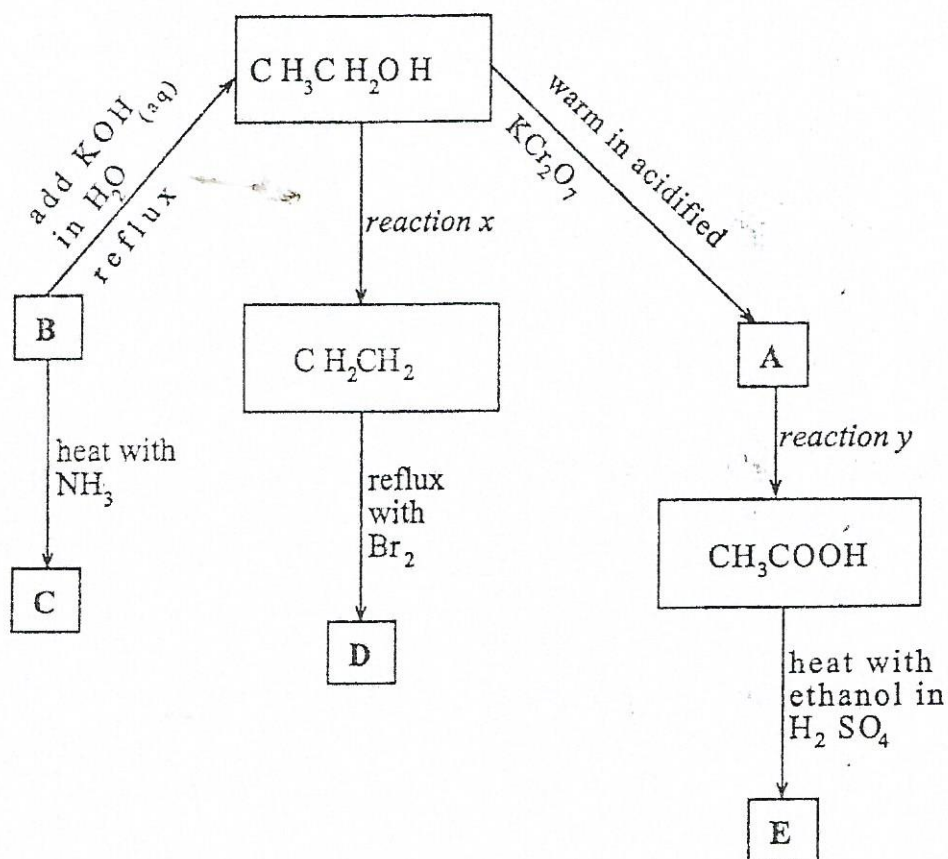


Fig.5.1

- (a) Complete Table 5.1 by stating the reagents and conditions for reactions x and y.

Table 5.1

reaction	reagent(s)	condition(s)
x	_____	_____
y	_____	_____

[4]

[Turn over

(b) Identify the possible molecular formula for

(i) A. _____

(ii) B. _____

(iii) C. _____ [3]

[Total: 7]